M.G.M's COLLEGE OF ENGINEERING, NANDED Date:28/09/2022

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

ASSIGNMENT NO.: 1(Academic Year 2022-23 ODD SEM)

SUBJECT: Discrete Mathematics CLASS:SY(CSE) - I & II

1. Define Proposition. Explain Logical Connectives in detail.
2. Let p and q be the proposition.

   p : I bought a lottery ticket this week.
   q : I won the million dollar jackpot on Friday.
   Express the following proposition into an English Sentences:
   a) $\neg p$ b) $p \vee q$ c) $p \wedge q$ d) $p \rightarrow q$ e) $p \leftrightarrow q$ f) $p \oplus q$
3. Explain Tautology and Contradiction with example.
4. Determine whether the expression is tautology or Contradiction:

   a). $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q))$ b). $(q \wedge \neg(p \rightarrow q))$
5. Show that $\neg (p \wedge q)$ and $(\neg p \vee \neg q)$ is Logically Equivalence.
6. Explain Converse, Contra-positive and Inverse. "If it Snows tonight, then I will stay at home". Determine Converse, Contra-positive and Inverse.
7. Explain Predicates and Quantifiers in details.
8. Let P(x) be the statement. " x spends more than five hours every weekday in class". where the domain of x is the set of students. Translate the following English sentence into Quantifiers :

   1) Some students do not spend more than five hours every weekday in class.
   2) All students spend more than five hours every weekday in class.
   3) There is at least one student who spends more than five hours every weekday in class.
   4) Every student does not spend more than five hours every weekday in class.
9. Define set. Explain different operation of sets with example.
10. Define: a) Equality b) Subset c) Empty set d) power set e) Cardinality of set f) proper subset.
11. Let A = { a,b,c,d,e,} and B = { a,b,c,d,e,f,g,h }.

    Find: 1) $A \cup B$ 2) $A \cap B$ 3) $A - B$ 4) $B - A$
12. Suppose that there are 1807 freshmen at your school. Of these 453 are taking a course in Computer Science, 567 are taking course in Mathematics and 299 are taking course in both Computer Science and Mathematics. How many are not taking course in either Computer Science or in Mathematics?.
13. Define Prime. Find Prime Factorizations of 100, 641, 999 and 1024.
14. Find gcd and lcm of (24, 36) and (12,9) ?

15. Find gcd(476, 152) using Euclidean Algorithm.

16. Use Mathematical induction to show that :

$1^2 + 2^2 + \text{---------} + n^2 = n(n + 1)(2n + 1) / 6$ whenever n is a positive integers.

17. Prove that : a) " if n is Odd integer then 5n+6 is Odd".
b) " if n is an integer and $n^3 + 5$ is Odd then n is Even ".

18. Define Function. Explain different types of function with example.

19. Let *f1 and f2* be the functions then find $f1 + f2$ *and* $f1.f2$ from R to R.

Such that : $f1(x) = x^2$ *and* $f2(x) = x - x^2$

20. Explain Composite and Inverse of Functions with example.

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Subject In-Charge